

# Channeling Creativity Through a Deeper Understanding of AI Image Generation

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## ABSTRACT

AI-generated images burst onto the scene about a year ago, with tools like Stable Diffusion, Midjourney, and DALL-E 2 all making their debut in 2022. How do these models work, and how can they be used in a production setting? In this talk, we will give an overview of how models like DALL-E 2 work and how to leverage their architectures to make them truly useful tools in the creative process. Although there are differences between each specific model architecture, the takeaways from understanding this particular stack are transferable to the others.

## ACM Reference Format:

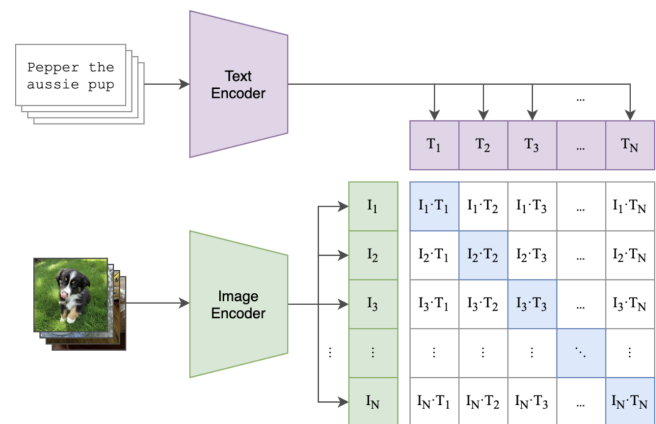
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## 1 LAB BRIEFING

In order to develop an intuition for how to coax the desired output from DALL-E 2, it is useful to understand how these generative models work. This model consists of three key components, a text encoder, a prior, and a decoder. The text encoder converts a text description into a vector, usually consisting of several hundred dimensions, that represents its semantic information. This is called an embedding in the latent space. The decoder does the inverse operation, producing an image based on an embedding in this space.

In the DALL-E 2 stack, we use CLIP - or Contrastive Language-Image Pretraining, as the text encoder. CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. Given a batch of  $N$  (image, text) pairs, CLIP is trained to predict which of the  $N \times N$  possible (image, text) pairings across a batch actually occurred. It does this by maximizing the cosine similarity of the image and text embeddings of the  $N$  real pairs in the batch while minimizing the cosine similarity of the embeddings of the  $N^2 - N$  incorrect pairings. Thus, CLIP embeddings are a learned representation of a shared text and image latent space, which can be used for a variety of purposes including semantic search, image deduplication, and image generation.

The next step is to take this text representation and use it to create representations of the possible output images, since a single



**Figure 1: CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples.**

text description can correspond to numerous different images. In order to allow the model to depict the range of images that might be derived from a single caption, we introduce another model we call the "prior". The main premise of the prior is to convert text information into an image embedding. It takes an image  $x$  and caption  $y$ , with embeddings  $z_i$  and  $z_t$  respectively, and is trained to produce CLIP image embeddings  $z_i$  conditioned on captions  $y$ . In addition to enabling a greater variety of outputs, this intermediate model also improves the quality of the image sample.

The decoder is the final step in the image generation process. This is a diffusion model that works by reversing the process of adding Gaussian noise to an image, and using a network to predict the denoised image at each step. It takes the embedding  $z_i$  output of the prior as a conditioning signal, and learns to recover the original image that matches the conditioning signal, while preserving the global and local coherence of the image. The model can be guided to generate images of different content and styles by varying the conditioning signal and the initial noise.

A more advanced application of the model for exploring the creative space is to use an existing image or images as the conditioning signal for the decoder. A single image can be embedded and then decoded to explore semantically similar images, or can be used to blend between two images by interpolating their CLIP image embeddings and then decoding with a diffusion model. The interstitial images can be novel or unexpected combinations, and reveal hidden patterns, connections, or similarities between seemingly different images. Text embeddings can be interpolated similarly.

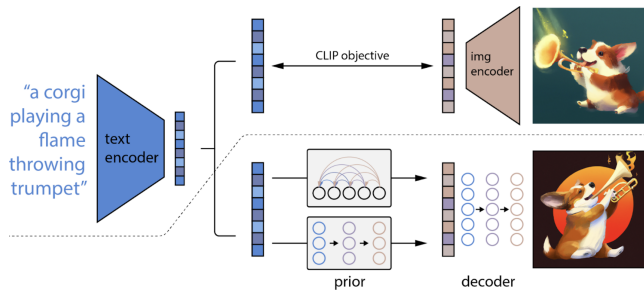
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**Figure 2: A CLIP text embedding is first fed to a prior to produce an image embedding, and then this embedding is used to condition a diffusion decoder which produces a final image.**



**Figure 3: Images generated with diffusion from embeddings along the interpolation between two source images.**

## 2 ATTENDEE EXPERIENCE

The session will begin with the introductory briefing to provide a basic understanding of how image generation models work. Following this, attendees can explore a few real world workflows that leverage the knobs available in this architecture via a browser-based sandbox provided for the session.

The concept stage of any project is where these new models offer a dramatic reduction in the time required to move from ideations to concept art. Concept artists can use the generated images as placeholders to communicate their vision, then refine them further with their own style and standards. We invite attendees to explore how AI-generated images can provide a fast and diverse source of inspiration and reference from nothing more than a handful of phrases or words. We will share a few examples workflows from artists-in-residence to demonstrate the diversity of methods and applications.

Another step in the creative process is iteration on details, when the rough idea has been achieved but there are elements to be dialed in. Through CLIP our image generation model has a fairly sophisticated understanding of natural language, though this understanding is still far from human level and still requires an artist in the loop to direct the output. We use the decoder to explore the latent space of a CLIP model through a variety of sampling parameters. These settings will be enabled in the sandbox to allow attendees to explore how these control can be exercised to nudge resulting images in the desired direction.

## 3 TAKE AWAY

By having a deeper understanding of how generative image models work, creators can develop an intuition for how to collaborate with these models to achieve inspiring results. Ultimately these models can be thought of as an exciting new addition to the suite of tools available to artists.

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